

2023-Ph-H1-Q5

Section: Particles and Waves

Topic: Interference

A beam of light passes through two narrow slits. Which pattern forms on a screen?

Answer: D

(Alternating bright and dark fringes)

Guidance for Students:

This is the classic **Young's double slit experiment**, where constructive and destructive interference leads to a fringe pattern.

Revision Tips:

- Bright fringes = constructive interference.
 - Dark fringes = destructive interference.
 - Pattern spacing depends on slit separation, wavelength, and distance to screen.
 - Interference proves wave nature of light.
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